

Machine Learning Approaches for Arabic NLP

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Abstract

This paper investigates transformer architectures for morphological disambiguation.

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1. Methodology and Model Architecture

We utilized bidirectional cross-attention with 24 layers and 16 attention heads.
Training was conducted on a multi-genre Arabic dataset comprising 50M tokens.

Table 1: Hyperparameter Configuration

Parameter	Layer Count	Learning Rate	Batch Size
Value	24	1e-4	128

2. Results and Conclusion

The proposed model achieved 94.8% F1-score across benchmark datasets.

المودل المقترح حقق 94.8% من F1-score عبر مجموعات الاختبار.

References

[1] Vaswani et al., Attention Is All You Need, NeurIPS 2017.
[2] Al-Badr et al., Arabic Semantic Benchmarks, ACL 2024.